

Allan Tsai

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Internship Experience

Tencent Lightspeed Studio - Game Client Programmer

July 2025 - Sep 2025

- **Overview:** Developing an unreleased game project in Unreal Engine 5, utilizing both C++ and Blueprints to implement diverse gameplay features.
- **Tools Usage:** Utilize different tools and plugins of Unreal Engine, including GAS (Game Ability System), Behavior Tree, Narrative 3, UMG, Common UI, etc.
- **Tools Made:** Developed data driven focused tools for the designers to easily configure based on needs, including Skill Table, Quest Nodes, and UI Displays.
- **Structure Designs:** Worked with colleagues and mentors to create a scalable design of the ability acquiring mechanics of the game, ensuring a flexible and decoupled design and integrating it with other components of the game such as the user interfaces and environment interactions.

Project Experience

Egaku - Unity Multiplayer Collaboration Game - Lead Programmer / Designer

- **Project Overview:** An online cooperative game where one player controls a runner to reach the goal, the other player acts as the painter, drawing platforms to help the character overcome obstacles.
- **Online Connection:** Use the Photon Pun SDK for lobby creation, data synchronization across the players, corresponding visual display and actions based on the player role.
- **Dynamic Mesh Platform Creation:** Use the mesh system in Unity to dynamically build out the platform along with the painter's mouse, with multiple pens that can be used by the player variant from pen size, mass, gravity on the object, interaction, etc.

Unity Rhythm Game – Personal Project

- **Project Overview:** This project aims to replicate the gameplay of the game Muse Dash, featuring two tracks for notes like tap notes and hold notes.
 - **Action Synchronization:** Notes are generated corresponding to the beat of the music to ensure that elements are precisely synchronized with the music.
 - **Data Decoupling and Conversion:** The chart creation process and note generation system are separated to ensure that the chart editor and runtime logic remain independent. When importing chart data, external formats are converted into game-recognizable data structures, enhancing flexibility in data importation.
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Slimey – Unity 2D Adventure Game Made – Lead Programmer

- **Project Overview:** Features a unique wall-climbing movement mechanic, allowing players to traverse along walls they touch. Players can absorb different environmental elements for combat and puzzle-solving.

- **Anti-gravity Movement Mechanics:** Players can move along walls they touch, including anti-gravity suspension, vertical wall climbing, and spherical navigation. The system dynamically adjusts attachment angles to match wall curvature, ensuring smooth and natural movement.

- **Object Pooling:** Extended Unity's built-in object pooling system to manage frequently spawned and destroyed objects, such as bullets and sound effects, improving memory utilization and performance.

- **UI Operations:** Designed UI logic for gameplay without mouse input, implementing customizable key bindings in settings to offer a personalized gameplay experience.

Choices – Unity Simulation Visual Novel Game – Lead Programmer

- **Project Overview:** Features a system where players' choices affect their stats, leading to different storylines and events, enhancing immersion of the game.

- **Dialogue System:** Dialogue is loaded from CSV files and can trigger various effects, such as character animations, variable changes, and choice restrictions.

- **Action & Stats Based Event System:** An event system triggered by player choices and stats, including dialogue, options, and visual changes. Events can be assigned priorities such as actions, locations, and time periods to ensure consistency with the player's current state.

Education

University of California, Irvine

Aug 2026 - May 2028

- Master of Science in Computer Science (Game Development) program

University of California, Irvine

Sep 2022 - June 2026

- Major in Game Design and Interactive Media - 3.858 GPA

- Minor in Information and Computer Science

Work Experience

Game Design Learning Assistant

Sep 2025 - Dec 2025

- Assisted in teaching game design and programming fundamentals using Unity and C# in the class GDIM 31 (Programming for Games) at University of California, Irvine.

- Guided students through the development process, encouraging creativity and problem-solving skills.

- Fostered on active learning by balancing instruction with hands-on creativity, asked questions to guide students to the solution instead of direct answers and instructions.

UCI Housing - Center Attendant

Sep 2023 - July 2025

- Work in a team of 40 people from different years and different backgrounds.

- Provide different types of service such as mailroom, fitness center, and housing office.

- Enhanced problem-solving, communication, and feedback integration skills.

Skills

- C#, C++, Python, Unity, Figma, Unreal Engine, Twine, RPG Maker, Lua